

(1001CJA10212082)

Test Pattern



CLASSROOM CONTACT PROGRAMME (Academic Session : 2020 - 2021)

JEE (Advanced)

SCORE - II

13-05-2021

JEE(Main + Advance) : ENTHUSIAST & LEADER (PHASE : SCORE-II)

Test Type : SCORE - II

Test Pattern : JEE (Advanced)

TEST DATE : 13-05-2021
ANSWER KEY
PAPER-1
PHYSICS

SEC-1	Q.	1	2	3	4	5	6		
	A.	B,D	C	B,C	A,C,D	A,B,C,D	A,C		
SEC-2	Q.	1	2	3	4				
	A.	A	B	C	D				
SEC-3	Q.	1	2	3	4	5	6	7	8
	A.	2.66	9.00	2.00	1.00	2.00	2.80	1.20	20.00

CHEMISTRY

SEC-1	Q.	1	2	3	4	5	6		
	A.	B,C	A,B,D	A,B,C,D	A,B,C,D	A,B,C	A,B,D		
SEC-2	Q.	1	2	3	4				
	A.	A	A	B	A				
SEC-3	Q.	1	2	3	4	5	6	7	8
	A.	3.00	200.00	448.00	2.00	5.00	6.00	6.00	223.40

MATHEMATICS

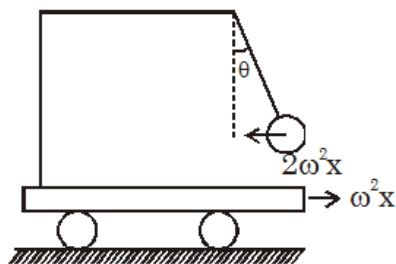
SEC-1	Q.	1	2	3	4	5	6		
	A.	B,C	C,D	B,C,D	A,B	A,B	A,D		
SEC-2	Q.	1	2	3	4				
	A.	A	B	A	C				
SEC-3	Q.	1	2	3	4	5	6	7	8
	A.	1.20	7.50	0.25	0.40	0.42, 0.43	1.45	4.66, 4.67	4.50

HINT – SHEET

PHYSICS

SEC-1

2. Ans (C)



$$1.8 \times \frac{\pi}{180} \times \ell \times x = 3A$$

$$A = \frac{\pi \ell}{150} = \frac{\pi}{150} \times \frac{3}{10} = \frac{\pi}{500} \text{ m}$$

Centre of mass is at rest.

$$3 \times x = 2l\theta$$

$$x = \frac{2l\theta}{3}$$

from carts frame

$$-(2g \ell \sin \theta + 2 \times \omega^2 x \times \ell) = 2\ell^2 \alpha$$

$$-(2g \ell \theta + 2 \times \omega^2 \ell \times 2 \frac{\ell \theta}{3}) = -2\ell^2 \times \omega^2 \theta$$

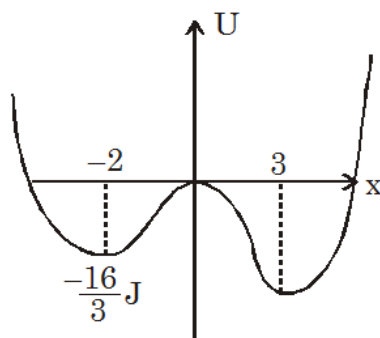
$$+ 2g \ell + \frac{4}{3} \omega^2 \ell^2 = 2\omega^2 \ell^2$$

$$2g \ell = \frac{2}{3} \omega^2 \ell^2$$

$$\omega = \sqrt{\frac{3g}{\ell}} \Rightarrow T = 2\pi \sqrt{\frac{\ell}{3g}}$$

$$2\pi \sqrt{\frac{0.3}{300}} = \frac{\pi}{5} \text{ sec}$$

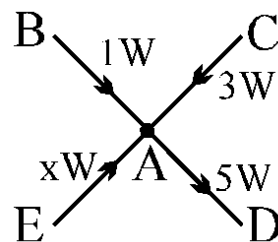
3. Ans (B,C)



To reach $x = 3$, it has to cross $x = 0$. Energy required = $\frac{16}{3} \text{ J}$

4. Ans (A,C,D)

Heat flow



At A

$$x + 1 + 3 = 5$$

$$x = 1$$

heat in flows from E

$$\therefore T_E > T_A$$

$$\therefore T_C > T_A > T_D$$

$$T_B - T_A = T_E - T_A$$

$$\therefore T_B = T_E$$

5. Ans (A,B,C,D)

$$R = v_x t$$

$$t = \sqrt{\frac{2(1-y)}{g}}$$

$$v_x = \sqrt{2gy}$$

$$\Rightarrow 0.8 = 2 \sqrt{y(1-y)}$$

$$0.4^2 = y(1-y)$$

$$y^2 - y + 0.16 = 0$$

$$y = \frac{1 \pm \sqrt{1-0.64}}{2} = \frac{1 \pm 0.6}{2} = 0.8 \text{ m or } 0.2 \text{ m}$$

6. Ans (A,C)

$$I_0 = 4I_0 \cos^2 \frac{\pi y d}{\lambda D}$$

$$\cos \frac{\pi y d}{\lambda D} = \pm \frac{1}{2}$$

$$\frac{\pi y d}{\lambda D} = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \dots$$

$$y = \frac{\beta}{3}, \frac{2\beta}{3}, \frac{4\beta}{3}, \dots$$

$$\Delta y = \frac{\beta}{3} \text{ or } \beta \text{ or } \frac{2\beta}{3} \text{ or } \frac{4\beta}{3}$$

$$\Delta y = (\mu - 1) \frac{tD}{d} = \frac{\pi}{3} \frac{\lambda D}{d}$$

$$\Rightarrow t = \frac{2n\lambda}{3} = 300 \text{ nm} \times n$$

PHYSICS

SEC-2

1. Ans (A)

Fundamental idea of electric & magnetic forces on charged particles.

$$\vec{F}_e = q\vec{E}$$

$$\vec{F}_m = q(\vec{V} \times \vec{B})$$

2. Ans (B)

$$\left. \begin{array}{l} v > 0 \text{ and } a > 0 \\ v < 0 \text{ and } a < 0 \end{array} \right\} \rightarrow \text{speeding up } \uparrow$$

$$\left. \begin{array}{l} v > 0 \text{ and } a < 0 \\ v < 0 \text{ and } a > 0 \end{array} \right\} \rightarrow \text{speeding down } \downarrow$$

$$(P) v = \frac{dx}{dt}, a = \frac{dv}{dt}$$

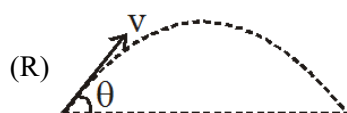
$$\text{at } t = 0 \Rightarrow v > 0 \text{ and } a < 0$$

$$(Q) x = t^2 + 3t$$

$$v = \frac{dx}{dt} = 2t + 3 \text{ at } t = 0 \Rightarrow v = 3 \text{ m/s}$$

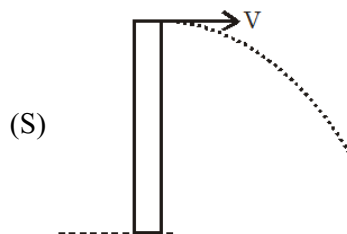
$$a = \frac{dv}{dt} = 2 \text{ (remain constant)}$$

$$\text{at } t = 0 \Rightarrow v > 0 \text{ and } a > 0$$



$$a < 0 \text{ at } t = 0$$

$$v > 0$$



$$a < 0 \text{ } t > 0$$

$$v < 0$$

3. Ans (C)

(A) $B \rightarrow C$ is isobaric expansion

$$T_B < T_C \text{ and } V_B < V_C$$

$$\text{So } \Delta U > 0 \text{ and } \Delta W > 0; \text{ Hence } \Delta Q > 0$$

(B) $A \rightarrow B$ is isochoric heating

$$T_B > T_C \text{ and } V_B = V_A$$

$$\text{So } \Delta U > 0 \text{ and } \Delta W = 0; \text{ Hence } \Delta Q > 0$$

(C) $A \rightarrow B \rightarrow C \rightarrow A$ is cyclic process

$$\Delta T = 0 \text{ So, } \Delta U = 0 \text{ But } \Delta W > 0 \text{ and Hence } \Delta Q > 0$$

(D) $C \rightarrow A$

$$T_A < T_C \text{ and } V_A < V_C$$

$$\text{So } \Delta U < 0 \text{ and } \Delta W < 0, \text{ So } \Delta Q < 0$$

4. Ans (D)

$$F = +20 \text{ cm}$$

$$u = -30 \text{ cm}$$

$$v =$$

$$\frac{uF}{u+F} = \frac{(-30)(20)}{-30+20} = \frac{+600}{-10} = +60 \text{ cm}$$

$$m = \frac{v}{u} = \frac{60}{-30} = -2$$

Real, inverted and magnified

$$(Q) F = -40 \text{ cm}$$

$$u = -30 \text{ cm}$$

$$v = \frac{(-30)(-40)}{-30+40} = 120 \text{ cm}$$

$$m = -\left[\frac{v}{u}\right] = -\left[\frac{120}{-30}\right] = +4$$

virtual, erect and magnified

$$(R) \frac{u_2}{v} - \frac{u_1}{u} = \frac{u_2 - u_1}{R}$$

$$\frac{1}{v} - \frac{3/2}{-20} = \frac{1 - 3/2}{-10}$$

$$\frac{1}{v} + \frac{3}{40} = \frac{1}{20}$$

$$\frac{1}{v} = \frac{1}{20} - \frac{3}{40} = \frac{2-3}{40} = -\frac{1}{40} \Rightarrow v = -40$$

$$m = +\frac{u_i}{u_r} \frac{v}{u}$$

$$= \frac{+3/2}{1} \times \left[\frac{-40}{-20}\right] = +3$$

virtual, erect, magnified.

$$(S) v =$$

$$\frac{uF}{u+F} = \frac{(-40)(-40)}{-40-40} = \frac{1600}{-80} = -20$$

$$m = \frac{v}{u} = \frac{-20}{-40} = +\frac{1}{2}$$

virtual, erect, diminished.

PHYSICS

SEC-3

2. **Ans (9.00)**

$$\phi_{\text{net}} = \frac{q_{\text{inside}}}{\epsilon_0} = \frac{6\lambda}{\epsilon_0}$$

$$\text{for } I_{\text{octant}} = \frac{6\lambda}{8\epsilon_0} = \frac{3\lambda}{4\epsilon_0}$$

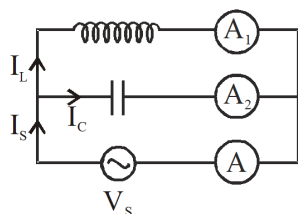
4. **Ans (1.00)**

$$\frac{p^2}{2m} = \frac{hc}{\lambda} - \phi \approx \frac{hc}{\lambda} = \frac{h^2}{2m\lambda^2}$$

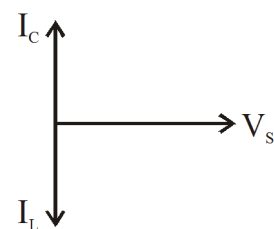
$$\lambda = \frac{h}{2mc} = \frac{6.60 \times 10^{-34}}{2 \times 9.1 \times 10^{-31} \times 3 \times 10^8}$$

$$= \frac{11}{91} \times 10^{-11} \text{ m} = 1.2 \text{ pm}$$

5. **Ans (2.00)**



Phasor diagram



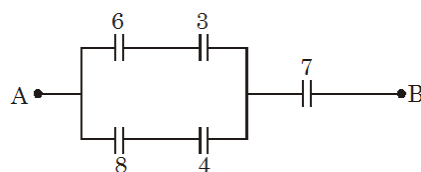
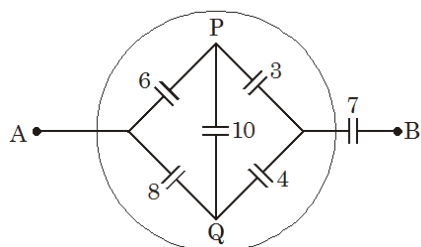
$$I_s = I_c + I_L$$

$$= 5 - 3$$

$$= 2 \text{ A}$$

6. **Ans (2.80)**

Balanced Wheat stone Bridge



$$C_{\text{eq}} = \frac{14}{5} \mu\text{F} = 2.80 \mu\text{F}$$

7. **Ans (1.20)**

$$N = N_0 e^{-\lambda t}$$

$$N_{\text{decays}} = N_0(1 - e^{-\lambda t})$$

$$= 10^{13} \left(1 - \frac{1}{4}\right) = \frac{3}{4} \times 10^{13}$$

$$Q = 1.6 \times 10^{-19} \times \frac{3}{4} \times 10^{13} = 1.2 \mu\text{C}$$

8. **Ans (20.00)**

$$10^3 \times a^3 \times g = 25 \text{ mg}$$

$$10^3 \times 0.8 a^2 y g + 10^3 a^2 (a - y) g = \text{mg}$$

$$\Rightarrow a^3 g = 25 + a^3 g - a^2 y g + 0.8 a^2 y g$$

$$10^3 \times 0.2 a^2 y g = 25$$

$$10^3 \times 0.2 a^3 \times 10 = 4$$

$$a^3 = \frac{8}{1000}$$

$$a = \frac{2}{10} = 0.2 \text{ m}$$

CHEMISTRY

SEC-3

1. **Ans (3.00)**

$$\ln \left[\frac{(t_{1/2})_1}{(t_{1/2})_2} \right] = \frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

$$\ln \left[\frac{3}{(t_{1/2})_2} \right] = \frac{840R}{R} \left(\frac{1}{300} - \frac{1}{400} \right)$$

$$(t_{1/2})_2 = 1.5 \text{ min.}$$

2. **Ans (200.00)**

$$W = -2(1-2)$$

$$= 200 \text{ J}$$

3. **Ans (448.00)**

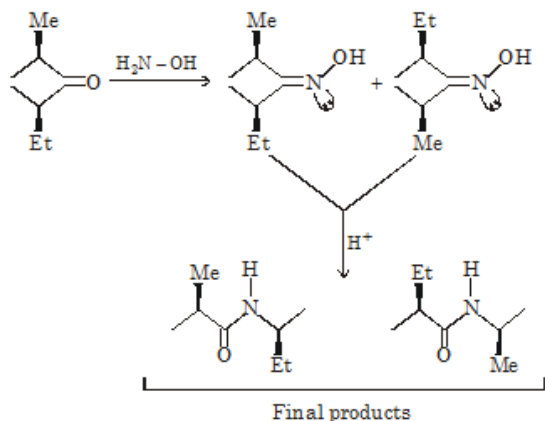
$$(0.4 + n) = \frac{2.8 \times 4}{0.0821 \times 273}$$

$$n = 0.1$$

$$\frac{0.4}{0.1} = \sqrt{\frac{M}{28}} \Rightarrow M = 448$$

4. **Ans (2.00)**

The configuration of migratory group remains unchanged



5. **Ans (5.00)**

(i), (ii), (iii), (vi), (viii)

MATHEMATICS

SEC-1

1. **Ans (B,C)**

$$\frac{x^4}{4} + \frac{y^2}{9} = 1 \Rightarrow e^2 = 1 - \frac{4}{9} = \frac{5}{9}$$

$$\frac{x^4}{4} - \frac{y^2}{9} = 1 \Rightarrow e^2 = 1 + \frac{9}{4} = \frac{13}{4}$$

$$3 < e_1^2 + e_2^2 < 4$$

2. **Ans (C,D)**

For $x \in [0, 1]$, $f'(x) \geq f'(1)$ (given condition in question)

$$\therefore \frac{f'(1)}{f^2(x) + 1} \leq \frac{f'(x)}{f^2(x) + 1}$$

On integrating we get,

$$f'(1) \cdot \int_0^1 \frac{dx}{f^2(x) + 1} \leq \tan^{-1}(f(1)) - \tan^{-1}(f(0))$$

$$\text{i.e. } \int_0^1 \frac{dx}{f^2(x) + 1} \leq \frac{\tan^{-1} f(1)}{f'(1)} \leq \frac{f(1)}{f'(1)}$$

$[\because \tan^{-1} x \leq x \text{ for } x \geq 0]$

For equality to hold, $\tan^{-1} f(1) = f(1)$

$$\text{i.e., } f(1) = 0 \text{ then } \int_0^1 \frac{dx}{f^2(x) + 1} = 0$$

which is not possible as this is a strictly

positive function. Hence, $\int_0^1 \frac{dx}{f^2(x) + 1} < \frac{f(1)}{f'(1)}$.

3. **Ans (B,C,D)**

Let $O(0,0,0)$, $A(0,0,2)$, $B(0,4,0)$, $C(6,0,0)$ P is point equidistant from zx, xy and yz plane.

Hence its coordinates are $P(r,r,r)$.

Equation of plane ABC is

$$2x + 3y + 6z - 12 = 0$$

$$r = \left| \frac{11r - 12}{\sqrt{49}} \right| \Rightarrow r = 3, \frac{2}{3}$$

$$\text{since } r < 2 \Rightarrow r = \frac{2}{3}$$

4. **Ans (A,B)**

$$A_k = \begin{bmatrix} C_{k-1} & 0 \\ 0 & C_k \end{bmatrix}, A_{k+1} = \begin{bmatrix} C_k & 0 \\ 0 & C_{k+1} \end{bmatrix}$$

$$A_k A_{k+1} = \begin{bmatrix} C_{k-1} C_k & 0 \\ 0 & C_k C_{k+1} \end{bmatrix}$$

$$\sum_{k=1}^{n-1} A_k A_{k+1} = \begin{bmatrix} \sum_{k=1}^{n-1} C_{k-1} C_k & 0 \\ 0 & \sum_{k=1}^{n-1} C_k C_{k+1} \end{bmatrix}$$

$$a = \sum_{k=1}^{n-1} C_{k-1} C_k = {}^{2n}C_{n-1} - n$$

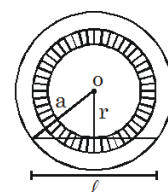
$$b = \sum_{k=1}^{n-1} C_k C_{k+1} = {}^{2n}C_{n+1} - n$$

$$a = b$$

$$a + b = 2({}^{2n}C_{n-1} - n)$$

5. **Ans (A,B)**

Let ℓ be the chord, r be the distance of mid point of chord from the centre of given circle with radius a .



$$\Rightarrow r^2 = a^2 - \frac{\ell^2}{4}$$

$$\ell \in \left(\frac{2a}{a}, \frac{6a}{4} \right), r \in \left(\frac{\sqrt{7}}{4}a, \frac{\sqrt{3}}{2}a \right)$$

The given condition is satisfied if mid point of chord is one of the points lying within the

circles of radii $\frac{\sqrt{7}a}{4}$ and $\frac{\sqrt{3}a}{2}$ concentric

with given circles.

The probability

$$= \frac{\text{The area of thin circular ring}}{\text{The area of circle}} = \frac{5}{16}$$

6. Ans (A, D)

The given equation is

$$acz_1z_2 + ad\bar{z}_1z_2 + bd\bar{z}_1\bar{z}_2 + bc\bar{z}_1z_2 \\ = acz_1z_2 + ad\bar{z}_1z_2 + bc\bar{z}_1\bar{z}_2 + bd\bar{z}_1\bar{z}_2$$

$$\text{i.e. } (ad - bc)(z_1\bar{z}_2 - \bar{z}_1z_2) = 0$$

$$\Rightarrow ad - bc = 0 \Rightarrow \frac{a}{b} = \frac{c}{d}$$

$$\text{If } z_1\bar{z}_2 - \bar{z}_1z_2 = 0$$

θ_1 and θ_2 are arguments of z_1, z_2

$$\Rightarrow \frac{|z_1|e^{i\theta_1}}{|z_1|e^{-i\theta_1}} = \frac{|z_2|e^{i\theta_2}}{|z_2|e^{-i\theta_2}} \Rightarrow \theta_1 = \theta_2$$

MATHEMATICS

SEC-2

1. Ans (A)

$$(P) \phi(x) = 3f\left(\frac{x^2}{3}\right) + f(3 - y^2)$$

$$\phi'(x) = 2x \left(f'\left(\frac{x^2}{3}\right) - f'(3 - x^2) \right)$$

$$\text{Now } \frac{x^2}{3} - (3 - y^2) = \frac{4x^2}{3} - 3 > 0$$

$$\Rightarrow x < -\frac{3}{2}, x > \frac{3}{2}$$

$$\frac{4x^2}{3} - 3 < 0 \Rightarrow -\frac{3}{2} < x < \frac{3}{2}$$

$$f'\left(\frac{x^2}{3}\right) - f'(3 - x^2) \text{ is positive in}$$

$$\left(-3, -\frac{3}{2}\right) \text{ and } \left(\frac{3}{2}, 4\right) \text{ and negative in}$$

$$\left(-\frac{3}{2}, \frac{3}{2}\right) \text{ as } f'(x) \text{ is increasing } (\because f''(x) > 0)$$

$$\therefore \phi'(x) \text{ is negative in } \left(-3, -\frac{3}{2}\right), \left(0, \frac{3}{2}\right)$$

$$\text{i.e. decreasing in } \left[-3, -\frac{3}{2}\right], \left[0, \frac{3}{2}\right] \text{ and}$$

$$\text{increasing in } \left[\frac{3}{2}, 4\right], \left[-\frac{3}{2}, 0\right]$$

$$(Q) g(x) = (a^2 - 3a + 2)\cos x + (a - 1)x + \sin x$$

$$f'(x) = -\frac{1}{2}(a^2 - 3a + 2)\sin \frac{x}{2} + (a - 1)$$

$$= (a - 1) \left(1 - \frac{1}{2}(a - 2)\sin \frac{x}{2} \right)$$

$$f'(x) = 0 \text{ when } a = 1 \text{ or } \sin \frac{x}{2} = \frac{2}{a - 2}, a \neq 2$$

$$\therefore \left| \frac{2}{a - 2} \right| < 1 \Rightarrow a \in (-\infty, 0) \cup (4, \infty)$$

$\Rightarrow$ It possess critical points
if $a \in (-\infty, 0] \cup [4, \infty) \cup \{1\}$

2. Ans (B)

$$S : x^2 + y^2 - 2x - 4y + 1 = 0,$$

centre (1, 2), radius = 2

$$S' : x^2 + y^2 - 4x - 2y + 1 = 0,$$

centre (2, 1), radius = 2

(P) (1, 3) lies inside S and outside S'.

$\Rightarrow$ Sum of tangents is 2.

(Q) Required circle is

$$x^2 + y^2 - 2x - 4y + 1 + \lambda(2x - 2y) = 0$$

$$x^2 + y^2 - 2(1 - \lambda)x - 2(2 + \lambda)y + 1 = 0$$

centre is $(1 - \lambda, 2 + \lambda)$

it lies on $2x + y - 6 = 0$

$$\Rightarrow 2 - 2\lambda + 2 + \lambda - 6 = 0$$

$$-\lambda - 2 = 0 \Rightarrow \lambda = -2$$

$$\text{radius} = \sqrt{9 + 0 - 1} = \sqrt{8}$$

$$\frac{r^2}{4} = 2$$

(R) $m = -1$

(S) Circles are intersecting

$\Rightarrow$ length of tangent

$$= \sqrt{(C_1C_2)^2 - (r_1 - r_2)^2} = 2$$

3. Ans (A)

$$(P) {}^{39}C_{3r-1} - {}^{39}C_{r^2} = {}^{39}C_{r^2-1} - {}^{39}C_{3r}$$

$${}^{40}C_{3r} = {}^{40}C_{r^2}$$

$$\Rightarrow (1) r = 3$$

$$(2) r^2 + 3r - 40 = 0 \Rightarrow r = 5$$

$$(Q) {}^{2n+3}C_{2n} - {}^{2n+2}C_{2n-1} = 15(2n + 1)$$

$${}^{2n+2}C_{2n} = 15(2n + 1)$$

$$(2n + 2)(2n + 1) = 30(2n + 1)$$

$$\Rightarrow n = 14$$

$$(R) {}^{56}P_{r+6} : {}^{54}P_{r+3} = 30800 : 1$$

$$\frac{56!}{(50 - r)!} \times \frac{(51 - r)!}{54!} = \frac{30800}{1}$$

$$56 \times 55 \times (51 - r) = 30800$$

$$51 - r = 10 \Rightarrow r = 41$$

$$(S) \binom{n+2}{8} : {}^{n-2}P_4 = 54 : 16$$

$$\frac{(n+2)!}{8!(n-6)!} \times \frac{(n-6)!}{(n-2)!} = \frac{57}{16}$$

$$\frac{(n+2)(n+1)(n)(n-1)}{8!} = \frac{57}{16}$$

$$(n+2)(n+1)(n)(n-1) = 143640$$

$$= 18 \times 19 \times 20 \times 21$$

$$\Rightarrow n = 19$$

4. Ans (C)

$$(P) \text{ H.M.} = 4 \Rightarrow \frac{2ab}{a+b} = 4$$

$$\Rightarrow ab = 2(a+b) \quad \dots(1)$$

$$A = \frac{a+b}{2}, G^2 = ab, 2A + G^2 = 27$$

$$\Rightarrow a+b+ab = 27$$

$$(1), (2) \Rightarrow a+b = 9, ab = 18$$

$$\Rightarrow a = 3, b = 6 \Rightarrow a^2 + b^2 = 45$$

$$(Q) a, x, y, z, b \text{ are in A.P.} \Rightarrow y = \frac{a+b}{2}$$

$x+z = a+d+b-d = a+b$, where d is the common difference.

$$\therefore x+y+z = 15 \Rightarrow \frac{a+b}{2} + (a+b) = 15$$

$$\Rightarrow a+b = 10 \quad \dots(1)$$

a, x, y, z, b are in H.P.

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = \frac{1}{2} \left(\frac{1}{a} + \frac{1}{b} \right) + \left(\frac{1}{a} + \frac{1}{b} \right)$$

$$= \frac{3}{2} \left(\frac{1}{a} + \frac{1}{b} \right)$$

$$= \frac{3}{2} \frac{(a+b)}{ab} = \frac{5}{8}, \text{ given}$$

$$\therefore ab = \frac{8}{5} \times \frac{3}{2} \times 10 = 24 \text{ [using (1)]} \quad \dots(2)$$

$$\text{Using (1) and (2)} \Rightarrow (a, b) = (4, 6)$$

$$\Rightarrow a^2 + b^2 = 52$$

$$(R) \sum_{r=1}^{16} \left(\frac{1^3 + 2^3 + \dots + r^3}{1 + 3 + \dots + (2r-1)} \right)$$

$$= \sum_{r=1}^{16} \frac{1}{4} (r+1)^2 = \frac{1}{4} (2^2 + 3^2 + \dots + 17^2)$$

$$= \frac{1}{4} \left(\frac{17 \times 18 \times 35}{6} - 1 \right) = 446$$

$$(S) G^2 = A^4$$

$$18^2 = A \cdot \frac{216}{13} \Rightarrow A = \frac{39}{2}$$

$$x_1 \text{ and } x_2 \text{ are roots of } x^2 - 2Ax + G^2 = 0$$

$$|x_1 - x_2| = \sqrt{(x_1 + x_2)^2 - 4x_1x_2}$$

$$= \sqrt{4A^2 - 4G^2} = 2\sqrt{A^2 - G^2} = 15$$

MATHEMATICS

SEC-3

1. Ans (1.20)

$$\tan \left(\frac{A-B}{2} \right) = \frac{a-b}{a+b} \cot \left(\frac{C}{2} \right) = \frac{1}{9} \cot \frac{C}{2}$$

$$\frac{31}{32} = \cos(A-B)$$

$$= \frac{1 - \tan^2 \left(\frac{A-B}{2} \right)}{1 + \tan^2 \left(\frac{A-B}{2} \right)} = \frac{1 - \frac{1}{81} \cot^2 \frac{C}{2}}{1 + \frac{1}{81} \cot^2 \frac{C}{2}}$$

$$\cot^2 \left(\frac{C}{2} \right) = \frac{81}{63} \Rightarrow \tan^2 \frac{C}{2} = \frac{7}{9} \Rightarrow$$

$$\cos C = \frac{1}{8}$$

$$\frac{a^2 + b^2 - c^2}{2ab} = \frac{1}{8} \Rightarrow c^2 = 36$$

2. Ans (7.50)

$$(1-x)^{30}(1+x+x^2)^{31}$$

$$= (1-x^3)^{30}(1+x+x^2)$$

$$\text{Coefficient of } x^{47} = \text{coeff. of } x^{45} \text{ in } (1-x^3)^{30}$$

$$= {}^{-30}C_{15}$$

3. Ans (0.25)

$$\text{Sum of the coefficients} = 0$$

$$\Rightarrow \text{one root is } 1$$

$$x^5 - 5x^4 + 9x^3 - 9x^2 + 5x - 1 = 0$$

$$(x+1)(x^4 - 4x^3 + 5x^2 - 4x + 1) = 0$$

$$x^2 - 4x + 5 - \frac{4}{x} + \frac{1}{x^2} = 0$$

$$\left(x + \frac{1}{x} \right)^2 - 2 - 4 \left(x + \frac{1}{x} \right) + 5 = 0$$

$$x + \frac{1}{x} = 1, 3$$

$$(1) x + \frac{1}{x} = 3 \Rightarrow x^2 - 3x + 1 = 0$$

$$x = \frac{3 \pm \sqrt{5}}{2}$$

$$(2) x + \frac{1}{x} = 1 \text{ (rejected)}$$

$$\text{real roots are } 1, \frac{3+\sqrt{5}}{2}, \frac{3-\sqrt{5}}{2}$$

4. Ans (0.40)

$$\text{Let } x = \frac{1}{t}$$

$$\lim_{t \rightarrow 0} \left(\frac{f(3+3t)}{f(3)} \right)^{\frac{1}{t}} = e^{\lim_{t \rightarrow 0} \left(\frac{f(3+3t) - f(3)}{t f(3)} \right)}$$

$$= e^{\lim_{t \rightarrow 0} \left(\frac{f'(3+3t) \cdot 3}{f(3)} \right)} = e$$

5. Ans (0.42, 0.43)

$$I = \int_{\pi/4}^{3\pi/4} \frac{\cos x}{1 - \cos x} dx$$

(k and A)

$$2I = \int_{\pi/4}^{3\pi/4} \cos x \left(\frac{1}{1 - \cos x} - \frac{1}{1 + \cos x} \right) dx$$

$$2I = 2 \int_{\pi/4}^{3\pi/4} \cot^2 x dx = 2 \left(2 - \frac{\pi}{2} \right)$$

$$I = 2 - \frac{\pi}{2}$$

6. Ans (1.45)

$$f(x) = \begin{vmatrix} \sec x & \cos x & \sec^2 x + \cot x \operatorname{cosec} x + \cos x \\ \cos^2 x & \cos^2 x & \operatorname{cosec}^2 x + \cos^4 x \\ 1 & \cos^2 x & \cos^2 x \end{vmatrix}$$

$$(R_1 \rightarrow R_1 - \sec x R_3, R_2 \rightarrow R_2 - \cos^2 x R_3)$$

$$f(x) = \begin{vmatrix} 0 & 0 & \sec^2 x + \cot x \operatorname{cosec} x \\ 0 & \cos^2 x - \cos^4 x & \operatorname{cosec}^2 x \\ 1 & \cos^2 x & \cos^2 x \end{vmatrix}$$

$$= (-\cos^2 x + \cos^4 x)(\sec^2 x + \cot x \operatorname{cosec} x)$$

$$= -\sin^2 x - \cos^3 x$$

$$\int_0^{\pi/2} f(x) dx = \int_0^{\pi/2} (-\sin^2 x - \cos^2 x) dx$$

$$= - \left(\frac{\pi}{4} + \frac{2}{3} \right)$$

7. Ans (4.66, 4.67)

x-coordinate of point of intersection is given by

$$4x^2 - 4(3+a)x + 9 + \frac{4a}{3} = 0$$

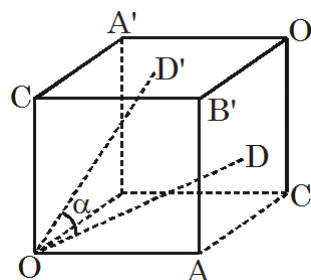
$$D = 0$$

$$(3+a)^2 - \left(9 + \frac{4a}{3} \right) = 0$$

$$a^2 + 6a - \frac{4a}{3} = 0$$

$$a^2 + \frac{14a}{3} = 0 \Rightarrow a = 0, -\frac{14}{3}$$

8. Ans (4.50)



$$\text{Let } |\vec{OA}| = 2$$

$$|\vec{OB}| = 4$$

$$|\vec{OC}| = 6$$

$$\vec{OD} = \vec{OA} + \frac{1}{2} \vec{AC'} + \frac{1}{2} \vec{AB'}$$

$$= 2\hat{i} + 2\hat{j} + 3\hat{k}$$

$$\vec{OD} = \vec{OC} + \frac{1}{2} \vec{CB'} + \frac{1}{2} \vec{B'O'}$$

$$= \hat{i} + 2\hat{j} + 6\hat{k}$$

$$\cos \alpha = \frac{\vec{OD} \cdot \vec{OD'}}{|\vec{OD}| |\vec{OD'}|} = \frac{24}{\sqrt{17.41}}$$